Derivation of the Indicial Equation

- Assume $y = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+r}$.
- Differentiate: $y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$.
- Differentiate again: $y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$
- Substitute y, y', y'' into $x^2 y'' + x[xP(x)]y' + [x^2 Q(x)]y = 0$.
- Collect the lowest power of x, namely x^r, coming from the $n = 0$ term.
- Its coefficient must vanish since $a_0 \neq 0$.
- This gives $[r(r-1) + p_0 r + q_0] a_0 = 0$.
- Since $a_0 \neq 0$, we obtain the **indicial equation**

$$r(r-1) + p_0 r + q_0 = 0.$$

- Here $p_0 = \lim_{x \to 0} xP(x)$ and $q_0 = \lim_{x \to 0} x^2 Q(x)$.
- This is a quadratic equation in r, giving two roots r_1, r_2.
- These roots determine the possible leading behaviour of the solutions.



The Indicial Equation and Its Roots

- Indicial equation: $r(r-1)+p_0 r+q_0=0$, i.e. $r^2+(p_0-1)r+q_0=0$.
- Let r_1 and r_2 be its two roots, with convention $r_1 \geq r_2$ (if real).
- Coefficients of the higher powers of x^{n+r} give a recurrence relation.
- This recurrence expresses a_n in terms of $a_0, a_1, \ldots, a_{n-1}$.
- The recurrence relation itself depends on r.
- Substituting $r=r_1$ (the larger root) always yields one valid series solution.
- Substituting $r=r_2$ may or may not yield a second independent series solution.
- Whether it works depends on how r_1 and r_2 are related.
- There are exactly three distinct possibilities for real roots.
- These are classified as Case I, Case II, and Case III.



Three Cases Based on the Indicial Roots

- Let r_1, r_2 be roots of the indicial equation with $r_1 \geq r_2$.
- **Case I:** $r_1 - r_2$ is not zero and not a positive integer.
- **Case II:** $r_1 = r_2$ (equal or repeated roots).
- **Case III:** $r_1 - r_2$ is a positive integer.
- In Case I, both roots give independent Frobenius series directly.
- In Case II, only one Frobenius series exists; a second solution needs a logarithm.
- In Case III, the smaller root may or may not give a second series.
- A logarithmic term may or may not appear in Case III as well.
- The precise form of the second solution is given by three theorems.
- These are stated in full on the next three slides.



Case I: Roots Differing by a Non-integer

- Suppose $r_1 - r_2$ is neither zero nor a positive integer.
- Then two linearly independent Frobenius solutions exist:

$$y_1 = x^{r_1} \sum_{n=0}^{\infty} a_n x^n, \qquad a_0 \neq 0,$$

$$y_2 = x^{r_2} \sum_{n=0}^{\infty} b_n x^n, \qquad b_0 \neq 0.$$

- Both series are obtained from the same recurrence relation.
- Substitute $r = r_1$ into the recurrence to compute the a_n's.
- Substitute $r = r_2$ into the recurrence to compute the b_n's.
- No logarithmic term is needed in either solution.
- General solution: $y = Ay_1(x) + By_2(x)$.
- This is the simplest of the three cases to handle.
- Problems 1 and 5 later in this lecture illustrate this case.



Case II: Equal (Repeated) Roots

- Suppose $r_1 = r_2 = r$ (a repeated root of the indicial equation).
- The first solution is the standard Frobenius series

$$y_1 = x^r \sum_{n=0}^{\infty} a_n x^n, \qquad a_0 \neq 0.$$

- A second independent solution necessarily contains a logarithm

$$y_2 = y_1(x) \ln x + x^r \sum_{n=1}^{\infty} b_n x^n.$$

- The logarithm arises because reduction of order produces a $\ln x$ term.
- The coefficients b_n are found by substituting y_2 into the ODE.
- Alternatively, b_n can be found by differentiating the recurrence with respect to r.
- General solution: $y = Ay_1(x) + By_2(x)$.
- This case always requires a logarithmic term; there is no exception. It arises in Bessel's equation of order zero.



Case III: Roots Differing by a Positive Integer

- Suppose $r_1 - r_2 = N$, a positive integer, with $r_1 > r_2$.
- The larger root always gives a valid Frobenius series

$$y_1 = x^{r_1} \sum_{n=0}^{\infty} a_n x^n, \qquad a_0 \neq 0.$$

- For the smaller root r_2, the recurrence may break down at $n = N$.
- If the coefficient of a_N becomes $0 \times a_N =$ (nonzero), a log term is forced.
- If instead it becomes $0 \times a_N = 0$, then a_N is arbitrary and no log is needed.
- The general form of the second solution is.

$$y_2 = C\, y_1(x) \ln x + x^{r_2} \sum_{n=0}^{\infty} b_n x^n, \qquad b_0 \neq 0$$

- Constant C may turn out to be zero (no log) or nonzero (log present) and both possibilities must be checked case by case.



Summary: Method of Solving by Frobenius Series

- Step 1: Verify $x = 0$ is a regular singular point (check $xP(x)$, $x^2Q(x)$ analytic).
- Step 2: Write the equation as $x^2y'' + x[xP(x)]y' + [x^2Q(x)]y = 0$.
- Step 3: Substitute $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ and its derivatives.
- Step 4: Collect the lowest power of x to obtain the indicial equation.
- Step 5: Solve the indicial equation for roots $r_1 \geq r_2$.
- Step 6: Identify which of Case I, II, III applies.
- Step 7: Derive the recurrence relation from the general coefficient of x^{n+r}.
- Step 8: Use $r = r_1$ in the recurrence to build the first solution y_1.
- Step 9: Build the second solution y_2 according to the applicable case.
- Step 10: Write the general solution $y = Ay_1 + By_2$.



Problem 1: Setup (Case I – Distinct Roots)

- Solve near $x = 0$: $\quad 2x^2y'' + xy' - (x+1)y = 0.$
- Here $P(x) = \dfrac{1}{2x}$, $Q(x) = -\dfrac{x+1}{2x^2}$
- Compute $xP(x) = \frac{1}{2}$ and $x^2Q(x) = -\frac{1}{2}(x+1)$, both analytic at $x = 0$.
- Hence $x = 0$ is a regular singular point.
- Assume the Frobenius series $y = \sum_{n=0}^{\infty} a_n x^{n+r}$.
- Then $y' = \sum (n+r)a_n x^{n+r-1}$,
  $y'' = \sum (n+r)(n+r-1)a_n x^{n+r-2}$.
- Substitute into $2x^2y'' + xy' - (x+1)y = 0$.
- Collect the x^{n+r} terms:

$$\sum_n \left[2(n+r)(n+r-1)+(n+r)-1\right]a_n x^{n+r} - \sum_n a_n x^{n+r+1} = 0.$$

- Lowest power ($n = 0$) gives the indicial equation

$$2r(r-1) + r - 1 = 0 \implies 2r^2 - r - 1 = 0.$$

- Factor: $(2r+1)(r-1) = 0 \implies r_1 = 1,\ r_2 = -\frac{1}{2}$.



Problem 1: Indicial Roots and Recurrence

- Roots found: $r_1 = 1$ and $r_2 = -\frac{1}{2}$.
- Difference $r_1 - r_2 = \frac{3}{2}$, which is not a positive integer.
- Hence this is **Case I**: two independent Frobenius series exist.
- General coefficient equation (from matching powers x^{n+r}):

$$[2(n+r)(n+r-1) + (n+r) - 1]\, a_n = a_{n-1}.$$

- Recurrence relation:

$$a_n = \frac{a_{n-1}}{2(n+r)(n+r-1) + (n+r) - 1} = \frac{a_{n-1}}{(2(n+r)+1)(n+r-1)}$$

- For $r = r_1 = 1$: $a_n = \dfrac{a_{n-1}}{(2n+3)n}$.
- For $r = r_2 = -\frac{1}{2}$: $a_n = \dfrac{a_{n-1}}{(2n)(n-\frac{3}{2})}$.
- Both recurrences are well defined for all $n \geq 1$ (no division by zero).
- This confirms both series exist without any obstruction.



Problem 1: Final Solution

- Using $r_1 = 1$: $a_1 = \dfrac{a_0}{5}$, $a_2 = \dfrac{a_1}{14}, \ldots$
- First solution: $y_1 = x\left(1 + \dfrac{x}{5} + \dfrac{x^2}{70} + \cdots\right)$
- Using $r_2 = -\frac{1}{2}$: $b_1 = \dfrac{b_0}{-1}$, $b_2 = \dfrac{b_1}{1}, \ldots$
- Second solution: $y_2 = x^{-1/2}\left(1 - x + x^2 - \cdots\right)$
- Both series converge for $x > 0$ with radius determined by the next singularity.
- Since $x = 0$ is the only singular point, both series converge for all $x > 0$.
- General solution: $y = A\,y_1(x) + B\,y_2(x)$ as

$$y = A\,x\left(1 + \tfrac{x}{5} + \cdots\right) + B\,x^{-1/2}\left(1 - x + x^2 - \cdots\right).$$



Example 2: Setup (Case I)

- Solve $4xy'' + 2y' + y = 0$ near $x = 0$.
- Standard form: $y'' + \dfrac{1}{2x}y' + \dfrac{1}{4x}y = 0$, so $P(x) = \frac{1}{2x}$, $Q(x) = \frac{1}{4x}$.
- Compute $xP(x) = \frac{1}{2}$ and $x^2Q(x) = \frac{x}{4}$, both analytic at $x = 0$.
- So $x = 0$ is a regular singular point, with $p_0 = \frac{1}{2}$, $q_0 = 0$.
- Assume $y = \sum_{m=0}^{\infty} a_m x^{m+r}$ and substitute into $4xy'' + 2y' + y = 0$.
- Collecting the coefficient of x^{m+r-1} for each m gives.

$$\big[4(m+r)(m+r-1) + 2(m+r)\big]a_m + a_{m-1} = 0, \quad a_{-1} = 0.$$

- At $m = 0$: indicial equation $4r(r-1) + 2r = 0 \implies 2r(2r-1) = 0$.
- Roots: $r_1 = \frac{1}{2}$, $r_2 = 0$; difference $\frac{1}{2}$ is not an integer, so **Case I** applies.



Example 2: Recurrence and Series

- General recurrence (for $m \geq 1$):
  $$a_m = \frac{-a_{m-1}}{2(m+r)\big(2(m+r)-1\big)}.$$
- **For $r = 0$:** $a_m = \dfrac{-a_{m-1}}{2m(2m-1)}$.
- $a_1 = -\frac{a_0}{2}$, $a_2 = \frac{a_0}{24}$, $a_3 = -\frac{a_0}{720}, \ldots$
- First solution: $y_1(x) = 1 - \dfrac{x}{2} + \dfrac{x^2}{24} - \dfrac{x^3}{720} + \cdots$
- **For $r = \frac{1}{2}$:** $a_m = \dfrac{-a_{m-1}}{(2m+1)(2m)}$.
- $a_1 = -\frac{a_0}{6}$, $a_2 = \frac{a_0}{120}, \ldots$.
- Second solution: $y_2(x) = x^{1/2}\left(1 - \dfrac{x}{6} + \dfrac{x^2}{120} - \cdots\right)$.
- We recognize these as familiar closed-form series.



Example 2: Final Solution

- Compare $y_1(x) = 1 - \frac{x}{2} + \frac{x^2}{24} - \frac{x^3}{720} + \cdots$ with the Maclaurin series of $\cos\sqrt{x}$.
- Indeed, $\cos\sqrt{x} = \sum_{k=0}^{\infty} \frac{(-1)^k x^k}{(2k)!} = 1 - \frac{x}{2!} + \frac{x^2}{4!} - \cdots$, an exact match.
- So $y_1(x) = \cos\sqrt{x}$.
- Compare $y_2(x) = x^{1/2}(1 - \frac{x}{6} + \frac{x^2}{120} - \cdots)$ with $\sin\sqrt{x} = \sqrt{x} - \frac{x^{3/2}}{6} + \frac{x^{5/2}}{120} - \cdots$
- So $y_2(x) = \sin\sqrt{x}$.
- General solution: $y(x) = C_1 \cos\sqrt{x} + C_2 \sin\sqrt{x}$
- This elegant closed form is exactly the result in the textbook : Kreyszig.
- It shows the Frobenius method can produce familiar elementary functions.



Problem 3: Setup (Case II – Equal Roots)

- Solve near $x = 0$: $xy'' + y' + xy = 0$ (Bessel's equation of order zero).
- Here $P(x) = \dfrac{1}{x}$, $Q(x) = 1$.
- Compute $xP(x) = 1$ and $x^2 Q(x) = x^2$, both analytic at $x = 0$.
- Hence $x = 0$ is a regular singular point.
- Assume $y = \sum_{n=0}^{\infty} a_n x^{n+r}$.
- Substitute into $xy'' + y' + xy = 0$ and collect powers of x^{n+r-1}.
- Lowest power ($n = 0$) gives the indicial equation

$$r(r-1) + r = 0 \implies r^2 = 0.$$

- Roots: $r_1 = r_2 = 0$ (a repeated root).
- Since $r_1 = r_2$, this is **Case II**.
- Only one Frobenius series exists directly; a logarithmic term will be required.



Problem 3: Recurrence and First Solution

- Matching the coefficient of x^{n+r-1} gives

$$(n+r)^2 a_n + a_{n-2} = 0 \quad (\text{with } a_{-1} = 0).$$

- Setting $r = 0$: recurrence becomes $n^2 a_n = -a_{n-2}$.
- Odd terms: $a_1 = 0 \Rightarrow$ all odd coefficients vanish.
- Even terms: $a_2 = -\dfrac{a_0}{4}$, $a_4 = \dfrac{a_0}{64}$, $a_6 = -\dfrac{a_0}{2304}, \ldots$
- General even coefficient: $a_{2k} = \dfrac{(-1)^k a_0}{2^{2k}(k!)^2}$.
- First solution (choosing $a_0 = 1$):

$$y_1(x) = J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k}(k!)^2} x^{2k}.$$

- This is the well-known **Bessel function of the first kind**, order zero.
- $J_0(x)$ converges for all x and is finite at $x = 0$.
- By Case II theory, the second solution must involve $\ln x$



Problem 3: Second Solution with Logarithm

- By Case II theory, the second solution has the form

$$y_2(x) = J_0(x) \ln x + \sum_{n=1}^{\infty} b_n x^n.$$

- Substituting into the ODE and matching coefficients (standard computation) gives that only even powers of x appear, as expected from the symmetry of the equation.
- The resulting coefficients satisfy $b_{2k} = \dfrac{(-1)^{k+1} H_k}{2^{2k}(k!)^2}$, where $H_k = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k}$ is the k-th harmonic number.
- This gives the standard second solution

$$y_2(x) = J_0(x) \ln x + \sum_{k=1}^{\infty} \frac{(-1)^{k+1} H_k}{2^{2k}(k!)^2} x^{2k}.$$

- A normalized version of y_2 is called the Bessel function of the second kind $Y_0(x)$.
- $Y_0(x) \to -\infty$ as $x \to 0^+$ due to the logarithmic term.

